

mapping process. First, two DNA sequences (fragments) are aligned. Then, to determine the accuracy of the alignment, the computer takes a random alignment of genes at any location, in any sequence, and assigns scores to each character-to-character comparison. The aligned pair with the highest score is then reported.

Another aspect of the use of computers in biology-related sciences is the use of information management systems, analysis tools and communication networks. Collectively known as bioinformation infrastructure, it incorporates bioinformatics (the use of huge databases of biological information) and computational biology (the use of algorithmic tools to facilitate biological analyses). Basically, bioinformation infrastructure bridges the gap between observations (the raw data), the analysis of this data to arrive at an understanding (information) of how the processes function and finally, the application (knowledge) of this information. Put simply, it helps us discover new, hitherto unknown relationships between data.

The tools

The study of the complex genetic structures that living organisms are made up of require specially developed software tools that would assist in the analysis of the large quantities of biological data. Special design tools are required to adapt commercial databases to meet the special needs of genome researchers. One such tool is called ERDraw, which is a graphical editing program for describing data in biological terms.

The scientists at Lawrence Berkley National Laboratory have been engaged in building large-scale data management systems, of which one of the first was the Chromosome Information System (CIS). It works in a manner similar to CAD (computer-aided design) software and lets biologists study chromosome maps and related data with the help of pictures and text.

In the CIS, data is easy to access and manipulate through an interface called GenomeGuide which is shareable through a management system called GenomeBase. Researchers can trace back information on a map to its experimental origins, and get more information about specific entries with this interface.

Other computer language based tools such as OML (Ontology Markup Language) are being developed by open source software groups, universities and software development firms to further the research and development on the biological fronts. Big Blue has been making supercomputers for this effect; the Department of Energy's Oak Ridge National Laboratory (ORNL) has acquired an IBM RS/6000 SP supercomputer system. The initial system, configured to perform 100 GigaOPps (1 billion calculations per second), will later be upgraded to 400 GigaOPps.

Computer scientists from ORNL and IBM are also working on developing new programming tools and techniques to support this highly sophisticated machine. They will also be expanding their collaborations in data storage and digital libraries to include parallel programming environments and Web-based supercomputing.

Compaq, Sandia National Laboratories and Celera Genomics are collaborating on developing system hardware and software solutions to address the demands of computational biology. The alliance will use Compaq Alpha processors—a family of RISC-based, 64-bit CPUs and computer systems—connected in a massively parallel configuration with extremely high bandwidth and low-latency mesh interconnects. The goal is to achieve 100 trillion operations per second (100 TeraOPps).

What next?

After the proper sequencing is complete and we get an actual map of DNA sequences, are we ready to retell the human story? Not quite. The quest doesn't end here; mapping the human genome is just the first step. Analysis of the human genome could help scientists in developing new medicines, crops, materials and solutions to challenges in energy development and environmental cleanup.

The information gleaned from HGP can be used to pinpoint the source of the dysfunctions that cause certain diseases, including cancer. This would lay the foundation for enormous advances in diagnosis and treatment of diseases.

Take, for example, the research on drugs. Many drugs used in treatment target the proteins at the cellular level. They bind with certain proteins to either enhance or inhibit their activity, thus controlling intracellular reactions.

The effectiveness of a drug depends on its ability to specifically bind to the target protein in order to minimise undesired side effects. Drug companies are hoping to employ the knowledge gained about the genetic code to predict which protein a particular gene sequence produces and which drugs or diagnostic tests may be effective.

Computers are playing an increasingly important part in helping researchers use models of the genetic, nucleic and other biological data to simulate the effect that new experimental drugs will have. Computers can help find out about possible biological functions of a protein by comparing it to databases of proteins with known function. Understanding the biological or biochemical mechanism of a disease often suggests the types of molecules needed for new drugs.

We still don't know what makes us tick, but computers can help us get mighty close to it. Till then, let's be content with what's in the genes. Us. 

DNA Computers and Us

The need for speed has prompted miniaturisation of the conventional silicon wafers used in computer processors and prompted researchers to look for materials that could surpass the existing speed limits.

Living organisms have millions of supercomputers—DNA molecules—in their body. DNA resembles a computer's hard drive in the way it stores data. Data transfer using DNA molecules is much faster than that in even the fastest supercomputers.

Adleman, a computer scientist, was the first to discover the computing potential of the DNA. He used DNA to solve the well-known Hamiltonian path problem. This revealed the computational and storage capabilities of DNA molecules in solving complex mathematical problems. Logic gates, which form the basis of all calculations in a computer, were formed using DNA molecules.

Developing microchips embedded with DNA instead of silicon poses many advantages: they are cheap, do not use toxic compounds, and are abundantly available. Furthermore, using DNA, miniaturisation of computer processors is much more feasible. Unlike the normal serial computing computers, DNA computers perform parallel computations. Soon computers with tens of gigabytes of speed would be available even for home users.